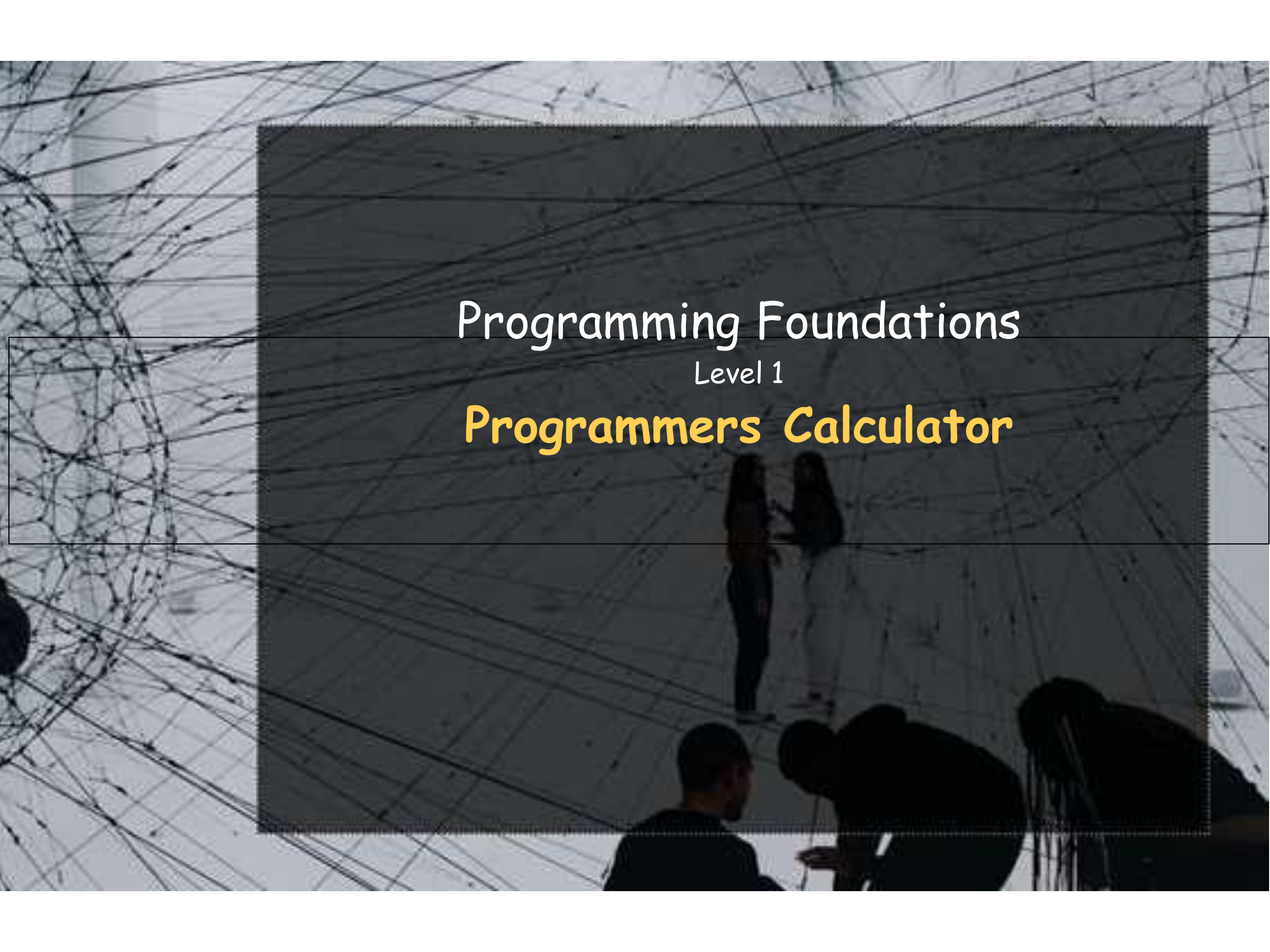




Programming Foundations Level 1

The background of the slide is a dark, abstract image. It features a dense, intricate network of thin, light-colored lines that resemble a complex web or a neural network. In the lower portion of the image, there are dark silhouettes of several people. Two people are standing in the center, facing each other, while others are seated or crouching in the foreground, looking towards the center. The overall tone is technical and modern.

Programming Foundations

Level 1

Programmers Calculator

Using the Programmer's Calculator to Convert Between Numbering Systems

Computers represent numbers in different **bases** (Decimal, Binary, Octal, Hexadecimal).

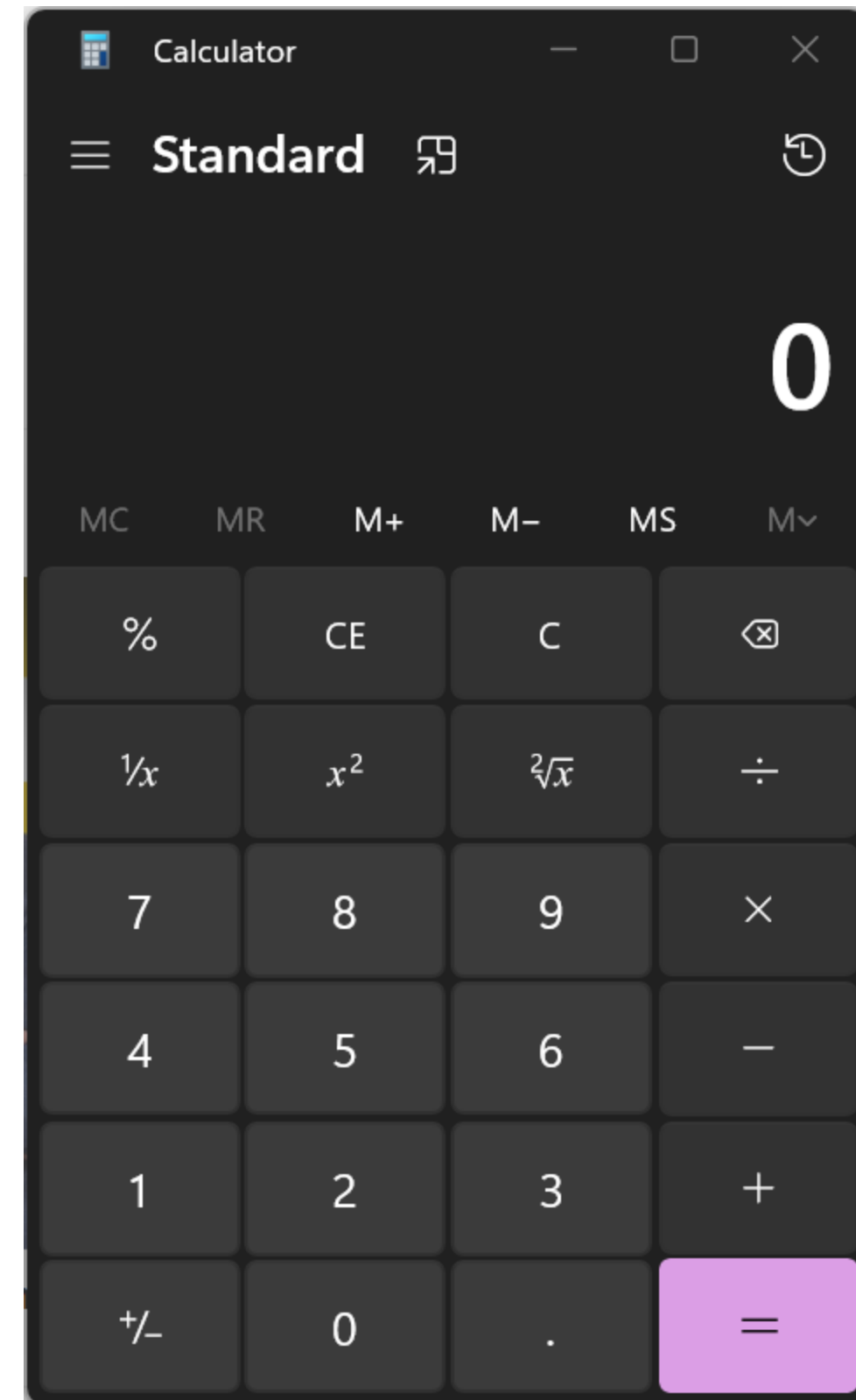
To quickly switch between them, developers and students use a tool built right into Windows — the **Programmer's Calculator**.

It allows you to:

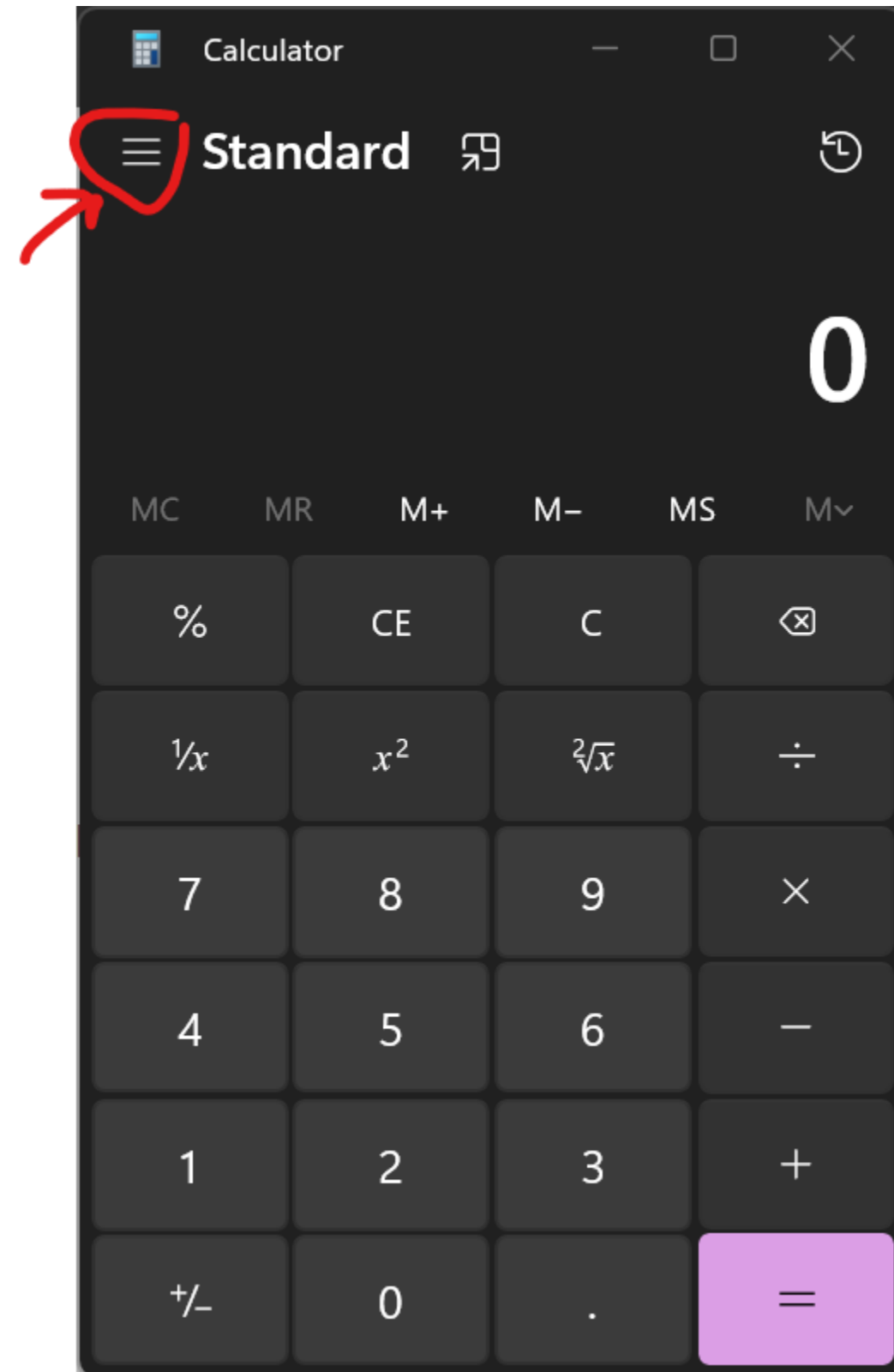
- ◆ Instantly convert values between **Binary (base-2)**, **Octal (base-8)**, **Decimal (base-10)**, and **Hexadecimal (base-16)**.
- ◆ See how the same value looks in each base.
- ◆ Avoid manual conversion errors while learning or coding.

Opening the Programmer's Calculator

-  **Steps:**
- 1 Press **Windows + R** → type **calc** → press **Enter**.



2 In the Calculator app, open the **menu** ($\equiv$) or click the **hamburger icon** on the top left.



3 Select **Programmer** mode (it may appear as "Programmer" or "Programmer's Calculator").

💡 Shortcut: Press **Alt + 3** (in newer Windows versions) to open Programmer mode directly.

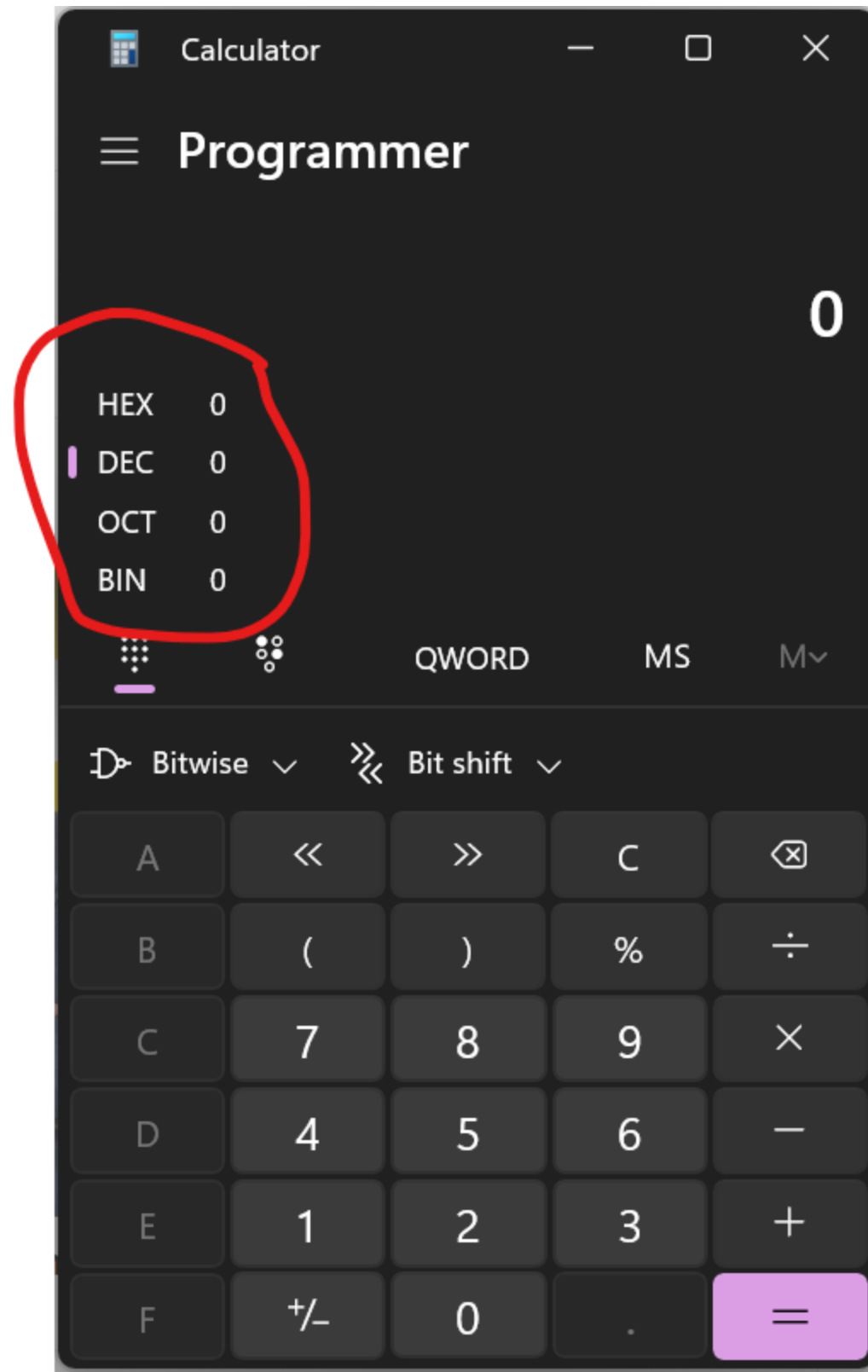
You'll now see four numbering systems on the left side of the window:

HEX → Hexadecimal (base 16)

DEC → Decimal (base 10)

OCT → Octal (base 8)

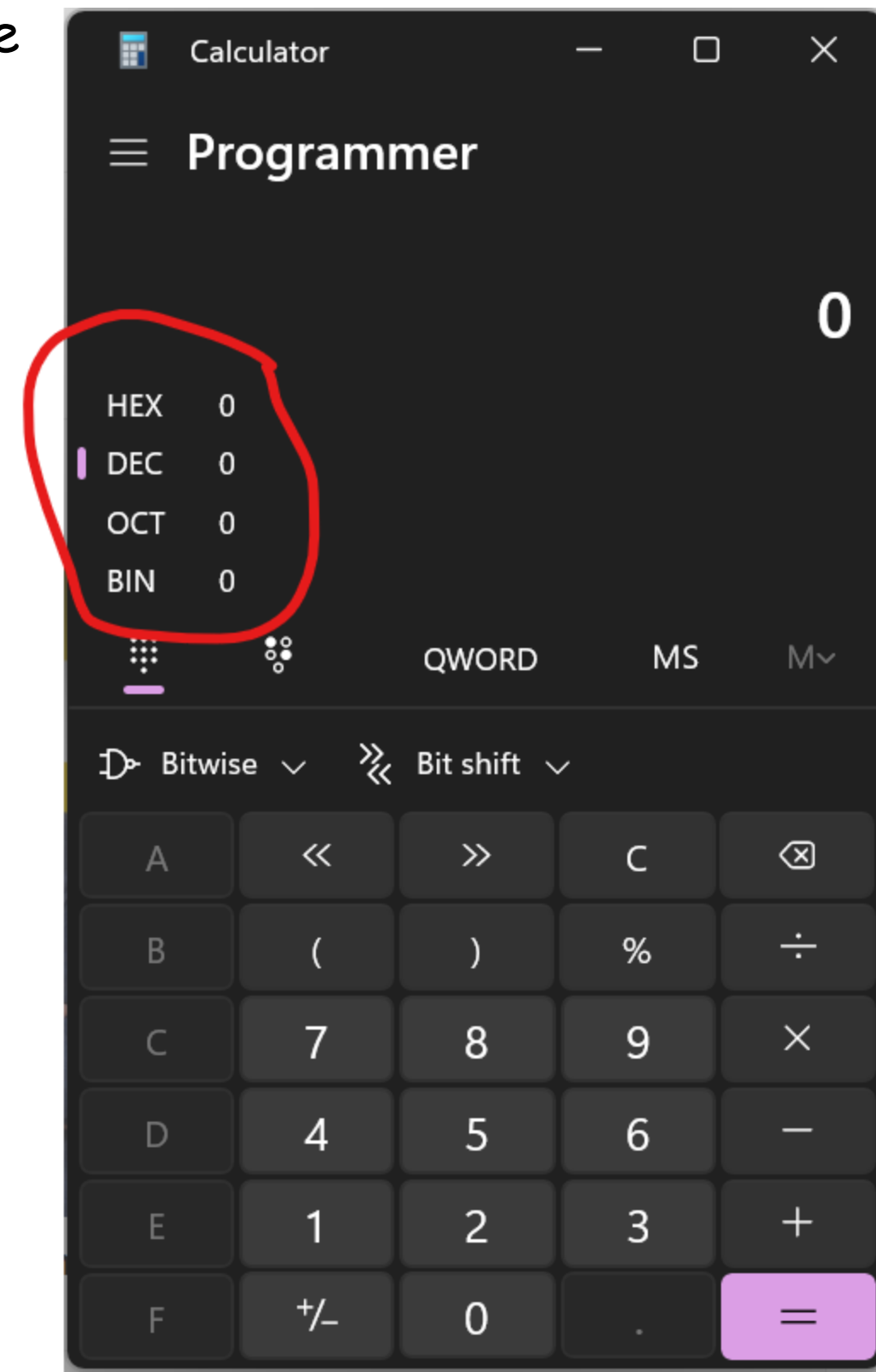
BIN → Binary (base 2)



◆ Understanding the Interface

The Programmer's Calculator shows:

- A numeric keypad (changes depending on the base you choose).
- Four buttons (HEX / DEC / OCT / BIN) to choose the input base.
- The corresponding values in all systems update automatically when you enter a number.

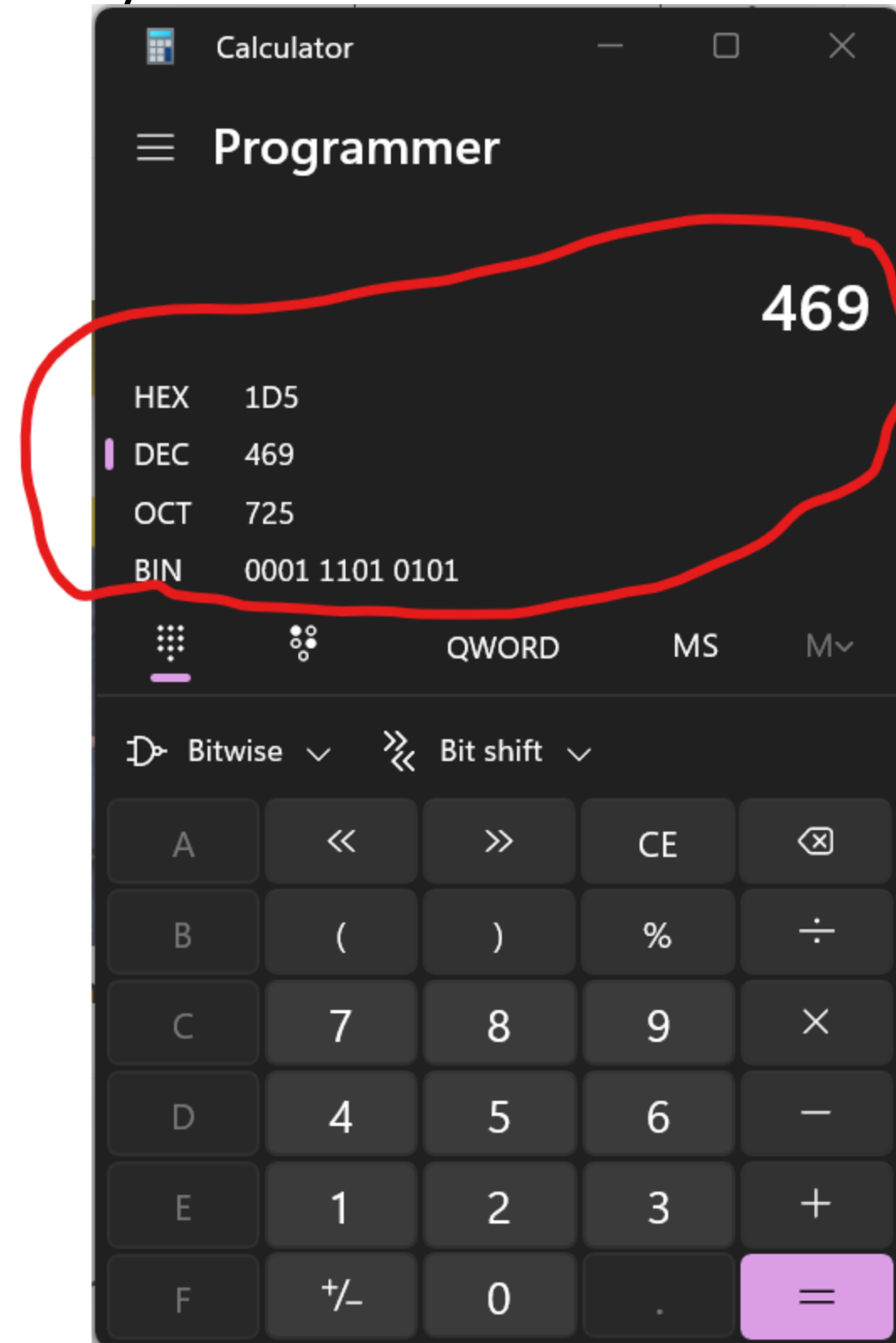


Example:

If you select **DEC**, type 469, the calculator will automatically show:

- HEX → 1D5
- OCT → 725
- BIN → 111010101

✓ It instantly converts across all four systems!



◆ Converting Decimal to Other Systems

 **Example: Convert 469 (decimal)**

① Select DEC (Decimal).

② Type 469.

③ Observe conversions:

HEX → 1D5

OCT → 725

BIN → 111010101

 **Explanation:** The calculator automatically interprets your input as Decimal and shows equivalent values in all other bases

◆ Converting Hexadecimal to Other Systems

📋 Example: Convert 1D5 (hex)

① Select HEX.

② Enter 1D5.

If your keyboard doesn't have letter keys active, ensure HEX is selected (then A-F buttons appear)

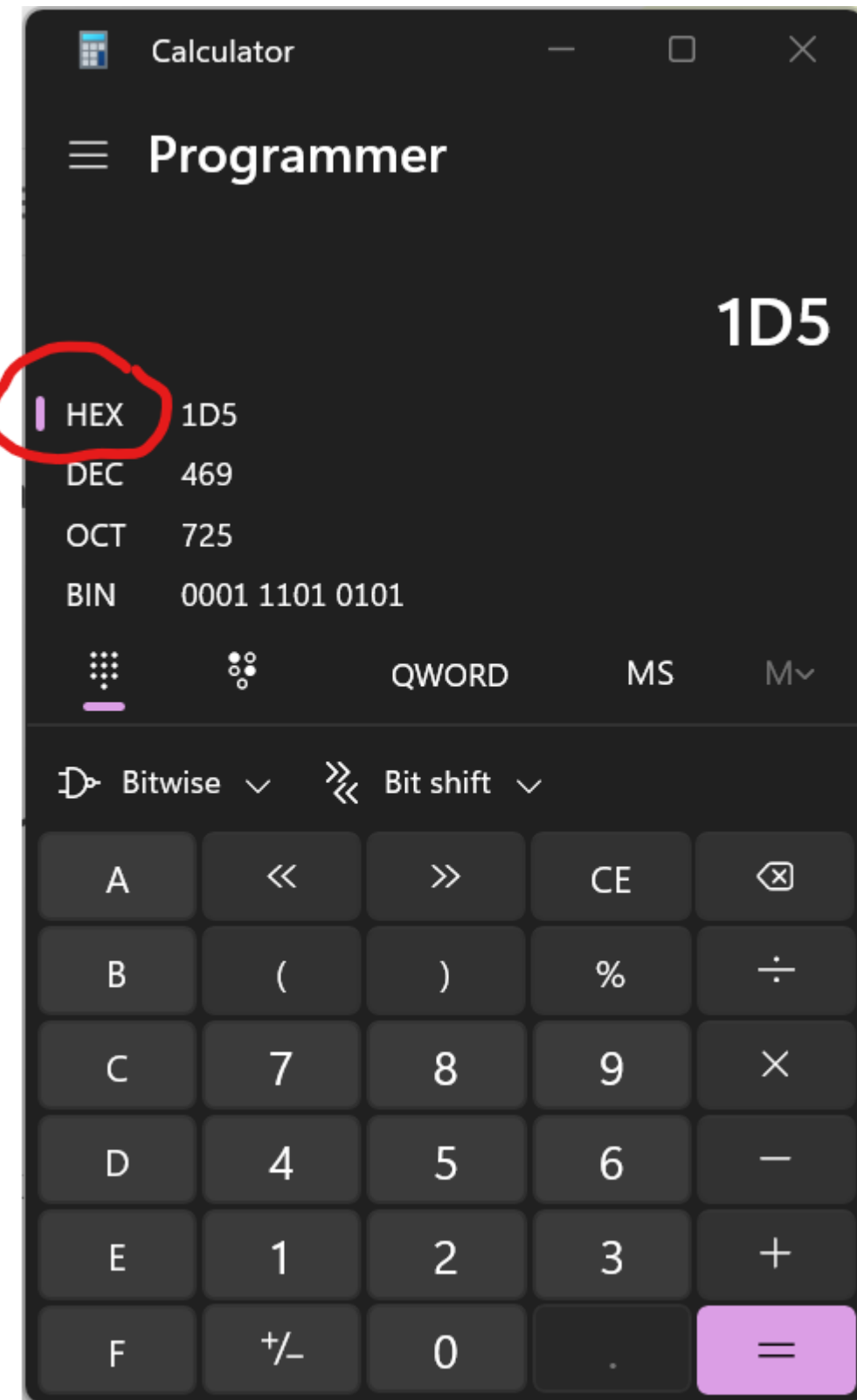
③ Observe:

DEC → 469

OCT → 725

BIN → 111010101

✅ The same logic applies — type in any base, and the others update instantly.



◆ Converting Binary to Other Systems

 Example: Convert 111010101 (binary)

- 1 Select **BIN**.
- 2 Type 111010101.
- 3 Observe:

OCT → 725

DEC → 469

HEX → 1D5

 Tip: The calculator only allows 0 and 1 keys when in binary mode preventing mistakes.

◆ Converting Octal to Other Systems

 Example: Convert 725 (octal)

① Select OCT.

② Enter 725.

③ Observe:

BIN → 111010101

DEC → 469

HEX → 1D5

 Notice: every conversion leads back to the same consistent value, regardless of starting base

◆ Using Word Size (for advanced learners)

In the Programmer's Calculator, you can change **Word size** (QWORD, DWORD, WORD, BYTE) to see how numbers behave when truncated (useful for programmers working with 8-bit, 16-bit, 32-bit, or 64-bit systems).

Example:

Change word size → select **BYTE** → notice that large numbers wrap around if they exceed 255 (since a byte = 8 bits).

Interconnection

- ◆ **Binary (base 2)** → machine-level representation (0s & 1s).
- ◆ **Octal (base 8)** → legacy system (3-bit groupings).
- ◆ **Decimal (base 10)** → human-readable.
- ◆ **Hexadecimal (base 16)** → compact developer shorthand (4-bit groupings).

By mastering the **Programmer's Calculator**, you can see instantly how all four systems interconnect — no manual conversion needed! 

Activity

Instructions:

Use the Windows Programmer's Calculator to perform the following conversions. Write your answers in **all four systems** (HEX, DEC, OCT, BIN).

- 1 Convert **255 (DEC)** to HEX, OCT, and BIN.
- 2 Convert **0xA3 (HEX)** to DEC, OCT, and BIN.
- 3 Convert **110101 (BIN)** to DEC, OCT, and HEX.
- 4 Convert **725 (OCT)** to DEC, BIN, and HEX.

Activity Solution

 Solution:

Input	HEX	DEC	OCT	BIN
255 (DEC)	FF	255	377	11111111
A3 (HEX)	A3	163	243	10100011
110101 (BIN)	35	53	65	110101
725 (OCT)	1D5	469	725	111010101

🏁 Conclusion

- ✅ The **Programmer's Calculator** is an essential tool for anyone learning **Binary, Octal, Decimal, and Hexadecimal systems**.
- ✅ It allows **instant cross-conversion** with zero risk of manual errors.
- ✅ It helps visualize **bit sizes, overflow, and representation** — key concepts for programmers.

By practicing with this tool, you'll quickly become fluent in reading and converting between numbering systems used in computer architecture and coding. 💻 ⚡

Thank you 😊